

A Case Study on Ranking of players in a tournament using matrix model

1. Problem Statement

Build comparison matrix and compute eigenvector to obtain relative rankings.

Given results of a tournament with n players, construct a comparison matrix representing match outcomes and compute:

- Principal eigenvalue
- Corresponding eigenvector
- Normalized eigenvector $\rightarrow$ Final ranking scores

Rank players based on descending eigenvector values.

2. Introduction

In competitive tournaments, ranking players purely based on the number of wins may not always reflect their true strength. A player defeating strong opponents should receive more weight than one defeating weaker opponent.

Matrix-based ranking methods use linear algebra and eigenvalues to compute relative strengths. This approach is similar to how Google PageRank works and is based on eigenvector centrality.

The ranking is obtained from the principal eigenvector of a comparison matrix.

3. Mathematical Model

Let there be n players.

Construct comparison matrix:

$$A = [a_{ij}]$$

Where:

$$a_{ij} = \begin{cases} 1 & \text{if player } i \text{ defeats player } j \\ 0 & \text{otherwise} \end{cases}$$

4. Eigenvector Ranking Equation

The ranking vector x satisfies:

$$Ax = \lambda x$$

Which can be written as:

$$(A - \lambda I)x = 0$$

Where:

- A = comparison matrix
- λ = eigenvalue
- x = ranking vector
- I = identity matrix

For non-trivial solution:

$$\det(A - \lambda I) = 0$$

Solve for eigenvalues, Choose:

$$\lambda_{max}$$

Then ranking vector:

$$x = \text{eigenvector corresponding to } \lambda_{max}$$

5. Example (4 Player based Tournament)

Given:

- P1 defeats P2, P3
- P2 defeats P3
- P3 defeats P4
- P4 defeats P1

Matrix:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Ranking equation:

$$Ax = \lambda x$$

Expanded system:

$$\begin{aligned}x_2 + x_3 &= \lambda x_1 \\x_3 &= \lambda x_2 \\x_4 &= \lambda x_3 \\x_1 &= \lambda x_4\end{aligned}$$

Solving gives eigenvalues and eigenvectors.

The eigenvector corresponding to largest eigenvalue gives ranking:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

We use raw values for ranking comparison.

6. Methodology

1. Collect match results
2. Form comparison matrix A
3. Compute eigenvalues and eigenvectors
4. Select principal eigenvector
5. Normalize vector
6. Sort values $\rightarrow$ Ranking order

This matrix formulation follows similar linear system representation as shown in your case study where matrix equation is formed as:

$$[A][X] = [B]$$

7. MATLAB Code

```
clc;
clear all;
close all;

n = 4;

A = [0 1 1 0;
     0 0 1 0;
     0 0 0 1;
     1 0 0 0];

disp('Comparison Matrix A = ');
disp(A);
```

```

[V,D] = eig(A);

eigenvalues = diag(D);

[max_eig, index] = max(real(eigenvalues));

ranking_vector = real(V(:,index));

disp('Largest Eigenvalue = ');
disp(max_eig);

disp('Raw Ranking Vector = ');
disp(ranking_vector);

[sorted_scores, order] = sort(ranking_vector,'descend');

disp('Final Ranking Order:');
for i = 1:n
    fprintf('Rank %d : Player %d (Score = %f)\n', ...
        i, order(i), sorted_scores(i));
end

figure;
bar(ranking_vector);
xlabel('Player Number');
ylabel('Eigenvector Value');
title('Tournament Ranking');
grid on;

```

